INTERNATIONAL MONETARY FUND

SAMOA

STAFF REPORT FOR THE 2021 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund (IMF) and the International Development Association (IDA)[1]

Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgment	*Yes. The projection horizon was extended to 20 years to capture the impact of natural disasters on debt dynamics.*

Samoa remains at high risk of external and overall debt distress, unchanged from the previous DSA published in April 2020. Consistent with the April 2020 DSA, judgement was applied to arrive at the high-risk rating, given that the mechanical external and overall risk rating from the model was assessed as moderate. To adequately capture Samoa's vulnerability to natural disasters and the effects of climate change, judgement was applied to extend the projection horizon to 20 years as opposed to the standard 10 years. This allows for the inclusion of the average long-term costs of natural disasters and climate change by incorporating their impacts on economic growth, the fiscal position, and the current account balance in the baseline scenario over the long term (FY2027-2041).[2] The combination of these two factors results in a breach of the threshold under the baseline scenario for the present value (PV) of the public debt-to-GDP ratio from 2034 onwards, as well as the PV of the external public and publicly guaranteed (PPG) debt-to-GDP ratio from 2038 onwards. The debt service-to-revenue ratio does not breach its indicative target because of the large share of concessional loans in external debt, and the temporary relief provided by the G-20 Debt Service Suspension Initiative (DSSI). A tailored natural disaster shock, similar in scale to the median impact of natural disasters in Samoa's history and assumed to strike in FY2022, leads to a further worsening of the debt dynamics. Given Samoa's vulnerability to natural disasters and external shocks, debt sustainability hinges upon the government's efforts to build fiscal and external buffers and enhance resilience. These efforts should focus on a mix of expenditure prioritization, and PFM reforms including improvements in budget planning, monitoring, and execution to help promote an inclusive economic recovery. Once the economy has recovered from the effects of COVID-19, revenue mobilization efforts should be strengthened. While domestic debt remains small, government-guaranteed debt accounted for 7.1 percent of GDP in FY2020, a reduction from FY2019.

[1] This DSA has been prepared jointly by the IMF and World Bank, following the revised LIC-DSF Framework and Guidance Note (2017) in effect as of July 1, 2018. The Samoa Composite Indicator (CI), which is calculated based on the October 2020 WEO and the World Bank's 2019 CPIA index, is 3.22, indicating that the country's debt-carrying capacity is strong. Thresholds for debt burden indicators are those established in the revised LIC-DSF framework.

[2] Consistent with previous analysis, see footnote 7. Samoa's fiscal year runs from July 1 to June 30.

PUBLIC DEBT COVERAGE

1. This Debt Sustainability Analysis (DSA) covers central government and central government-guaranteed debts (Text Table 1). Public and Publicly Guaranteed (PPG) debt accounted for 53.8 percent of GDP in FY2020, comprising central government debt (46.7 percent of GDP) and central government-guaranteed SOE debt (7.1 percent of GDP). There is no sub-national government structure in Samoa. The Central Bank of Samoa (CBS) is not allowed to contract debt on behalf of the government. The SOEs must seek government approval annually for all new loans. The authorities have begun reporting non-guaranteed SOE debt from FY2020 (estimated at around 1.1 percent of GDP in FY2020). This is not included in the debt definition but captured in the contingent liability stress test.[3] The definition of external and domestic debt is based on residency.[4] The authorities are currently working towards strengthening the monitoring and disclosing of fiscal risks with the assistance of PFTAC.

Text Table 1. Samoa: Debt Coverage

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	Not Applicable
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

BACKGROUND ON DEBT

2. Central government debt (referred to as public debt from here on) was 46.7 percent of GDP in FY2020, a decline of 0.8 percentage point of GDP from FY2019. An inadvertent tightening, driven by under execution of capital expenditure (CAPEX) due to implementation delays, led to the overall fiscal surplus of 6.2 percent of GDP in FY2020 compared to 2.7 percent of GDP in FY2019. Growth in total revenue, driven largely by an increase in grants, also contributed to the overall fiscal outcome. The surplus helped ensure timely repayments of external debt, thereby contributing to reducing the debt-to-GDP ratio.

3. Public external debt was US$375 million (46.3 percent of GDP) at the end of FY2020, a decline of 0.4 percentage point of GDP compared to FY2019 (Text Table 2). 54 percent of the outstanding public external debt at the end of FY2020 was from multilateral creditors, including the IDA and Asian Development Bank (AsDB), and 46 percent was from bilateral creditors (including China and Japan).

[3] The Government is strongly encouraged to report non-guaranteed SOE debt in the FY2022 budget documents and on an ongoing basis.

[4] Since all the domestic debt is in tala and all the external debt is in foreign currency, the residency-based classification is equivalent to the currency-based classification in the case of Samoa.

Text Table 2. Samoa: Evolution of Public Debt, FY2017 – FY2020

	FY2017	FY2018	FY2019	FY2020	FY2017	FY2018	FY2019	FY2020
	In millions of SAT	In millions of SAT	In millions of SAT	In millions of SAT	In percent of external debt	In percent of external debt	In percent of external debt	In percent of external debt
Total public debt	1047.4	1113.8	1058.6	1012.3	...	...	...	...
External	1018.0	1094.3	1043.7	1004.6	100.0	100.0	100.0	100.0
Multilateral	534.5	563.0	544.3	540.6	52.5	51.5	52.2	53.8
IDA	247.2	280.8	279.4	294.9	24.3	25.7	26.8	29.4
ADB	256.1	250.6	235.5	218.3	25.2	22.9	22.6	21.7
OPEC	23.2	24.0	22.6	20.9	2.3	2.2	2.2	2.1
Other	8.0	7.6	6.8	6.5	0.8	0.7	0.7	0.6
Bilateral	483.5	531.3	499.4	463.9	47.5	48.5	47.8	46.2
Exim Bank (China)	410.0	440.1	409.5	377.3	40.3	40.2	39.2	37.6
Japan	73.5	91.2	89.9	86.7	7.2	8.3	8.6	8.6
Domestic	29.4	19.5	14.9	7.8	...	...	...	...
Memorandum items:								
Nominal GDP (millions of SAT)	2110.0	2108.3	2231.2	2168.0				
Market rate (tala/U.S. dollar, end period)	2.5	2.6	2.6	2.7				

Sources: Samoan authorities; and IMF staff estimates

4. Domestic debt has been declining gradually and remains below 1 percent of GDP (Text Table 3). All domestic public debt is issued in local currency.

Text Table 3. Samoa: Stock of Public Debt at the End of FY2020

	In millions of SAT	In millions of US dollars	As a share of total debt	In percent of GDP
Total public debt	1012.3	377.9	100.0	46.7
External	1004.6	375.0	99.2	46.3
Multilateral	540.6	201.8	53.4	24.9
IDA	294.9	110.1	29.1	13.6
ADB	218.3	81.5	21.6	10.1
OPEC	20.9	7.8	2.1	1.0
Other	6.5	2.4	0.6	0.3
Bilateral	463.9	173.2	45.8	21.4
Exim Bank (China)	377.3	140.8	37.3	17.4
Japan	86.7	32.4	8.6	4.0
Domestic	7.8	2.9	0.8	0.4
Memorandum items:				
Nominal GDP (millions of SAT)	2168.0			
Market rate (tala/U.S. dollar, end period)	2.7			

Sources: Samoan authorities; and IMF staff estimates

5. Public and publicly guaranteed (PPG) debt totaled 53.8 percent of GDP in FY2020, of which guarantees were 7.1 percent of GDP. Government-guaranteed debt declined by 0.2 percentage point of GDP to 7.1 percent of GDP. More than half of the total guaranteed debt (about 54 percent in FY2020) originated from the Central Bank of Samoa (CBS) through the Credit Line Facility (CLF) to support SOEs (mainly DBS), following the aftermath of the natural disasters and promoting inclusive economic growth in

the targeted priority sectors.[5] Public financial institutions, including the Development Bank of Samoa (DBS)—received 96 percent of the total disbursements thus far—and the Samoa Housing Corporation (SHC), have engaged in on-lending operations, and have been making regular and timely repayments to the CBS. Including the loan through the CLF, operations by the DBS account for the largest share of government guarantees, holding around 65 percent of the total in FY2020 (Text Table 4).

Text Table 4. Breakdown of Government Guarantees

	FY2017	FY2018	FY2019	FY2020
By Creditor		In millions of Tala		
Unit Trust of Samoa	$ 10.7	$ 10.8	$ 25.619	$ 37.407
Bank of the South Pacific	$ 6.8	$ 5.1	$ 4.4	$ 4.0
Central Bank of Samoa	$ 99.1	$ 96.6	$ 95.7	$ 86.6
National Bank of Samoa	$ -	$ -	$ -	$ -
Accident Compensation Corporation	$ 17.0	$ 10.0	$ 10.0	$ 10.0
European Investment Bank	$ 16.6	$ 14.5	$ 11.3	$ 8.2
European Economic Committee	$ 0.2	$ 0.1	$ -	$ -
Samoa National Provident Fund	$ 17.6	$ 16.7	$ 15.4	$ 14.3
	$ 167.9	$ 153.9	$ 162.5	$ 160.5
By Borrower		In millions of Tala		
Samoa Shipping Services	$ 1.7	$ 1.7	$ 1.2	$ 1.1
Samoa Trust Estates Corporation	$ 4.2	$ 4.5	$ -	$ -
Samoa Housing Corporation	$ 14.2	$ 12.8	$ 11.5	$ 10.1
Polynesian Airlines/Samoa Airways	$ 6.8	$ 5.1	$ 24.3	$ 35.8
Development Bank of Samoa	$ 124.0	$ 119.9	$ 115.5	$ 103.4
Unit Trust of Samoa	$ 17.0	$ 10.0	$ 10.0	$ 10.0
Samoa Land Corporation	$ -	$ -	$ -	$ -
	$ 167.9	$ 153.9	$ 162.5	$ 160.5

Source: Authorities Data

COUNTRY CLASSIFICATION

6. Samoa's debt-carrying capacity remains strong (Text Table 5). The Composite Indicator (CI) index, which has been calculated based on the October 2020 WEO and the World Bank's 2019 Country Policy and Institutional Assessment (CPIA), is 3.22, indicating that the country's debt-carrying capacity remains strong in the LIC-DSA framework. The CI index was 3.29 in the April 2020 DSA.

[5] Based on the CBS Annual Report FY2018, the CBS has extended five separate CLF's since 2009, including the Tsunami CLF (2009), Cyclone Evan CLF (2013), and other CLFs to help promote inclusive economic growth in the priority sectors. The total disbursements amount to SAT 135 million, with the outstanding balance declining to SAT 96 million in FY2019 and SAT 87 million in FY2020.

Text Table 5. Samoa: Country Policy and Institutional Assessment Rating

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.985	1.53	48%
Real growth rate (in percent)	2.719	1.522	0.04	1%
Import coverage of reserves (in percent)	4.052	35.455	1.44	45%
Import coverage of reserves^2 (in percent)	-3.990	12.571	-0.50	-16%
Remittances (in percent)	2.022	15.494	0.31	10%
World economic growth (in percent)	13.520	2.928	0.40	12%
CI Score			3.22	100%
CI rating			Strong	

7. Based on the CI rating, Samoa's debt is assessed against a higher threshold than other PICs (Text Table 6). The relevant indicative thresholds for countries with a strong CI rating are 55 percent for the PV of PPG external debt-to-GDP ratio, 240 percent for the PV of PPG external debt-to-exports ratio, 21 percent for the PPG external debt service-to-exports ratio, and 23 percent for the external debt service-to-revenue ratio. The benchmark for the PV of total public debt under strong capacity is 70 percent.

Text Table 6. Samoa: Debt Thresholds

Debt carrying capacity (CI classification)	PV of PPG external debt in percent of		PPG external debt service in percent of		PV of total public debt
	GDP	Exports	Exports	Revenue	GDP
Weak	30	140	10	14	35
Medium	40	180	15	18	55
Strong	55	240	21	23	70

METHODOLOGY AND ASSUMPTIONS

A. The Baseline

8. The underlying baseline assumptions relating to the natural disaster stress test in the 2021 DSA follow the assumptions in the April 2020 DSA (Text Table 7). They are consistent with the macroeconomic framework based on updated data provided by the authorities and estimates by staff. Given Samoa's vulnerability to natural disasters and the effects of climate change, the baseline scenario incorporates the effects of natural disasters and climate change. The medium-term (FY2021–26) is assumed to be disaster free to simplify the policy discussion of the medium-term outlook. From FY2027 onwards, the baseline is adjusted to reflect the country's historical experience and incorporates the average long-term effects of natural disasters and climate change. It lowers GDP growth by 1.3 percentage points (pp), raises the current account deficit by 3.5 pp of GDP, and increases the fiscal deficit by 3.5 pp of GDP on

average each year during FY2027-41 vis-à-vis disaster-free projections during FY2021-26.[6] The assumptions are consistent with the findings of staff's analysis on the impact of natural disasters.[7] The discount rate used to calculate the net present value of external debt remains at 5 percent. The main assumptions are:

- **The economy is expected to severely contract in FY2021 and rebound modestly in FY2022**. Given the ongoing COVID-19 pandemic, significant declines in tourism earnings (with zero arrivals in FY2021) are assumed in the baseline, which would generate knock-on effects to non-tourism sectors and affect economic activity more broadly. In addition, floods around the time of the festive season dealt yet another blow to the economy, costing (based on the authorities' estimate) at least 1½ percent of GDP. Nonetheless, reconstruction efforts are expected to boost construction activity in the second half of FY2021. Also, resilient remittance inflows and the lifting of most social restrictions in December boosted household consumption. Moreover, a supplementary budget in December 2020 extending stimulus measures and a push by the government to fully execute its budget in FY2021 will likely promote economic activity during the second half of FY2021. Considering these developments, staff projects real GDP to decline by 7.8 percent in FY2021, rebounding to growth of 1.7 percent in FY2022, supported by tourism sector recovery and implementation of capital projects. Growth is projected to normalize at around 2.1 percent by the end of the medium term (FY2026). Real GDP growth during FY2027-31 period is lower due to the scarring effect of the pandemic. However, higher longer term growth is projected during the ensuing decade (FY2032-41), benefitting from implementation of structural reforms, including through the reform strategy of "Samoa 2040".[8] When the average impact of natural disasters is taking into account, GDP growth is lowered by 1.3 pp of GDP on average during FY2027-41.

- **Inflation is expected to remain contained.** Inflation is projected to fall to -2.5 percent in FY2021. Declines in global commodity prices and domestic utility prices (resulting from the fiscal stimulus) underpin staff's inflation projection for FY2021. Over the medium term, staff projects

[6] Calculation is done by subtracting the average of FY2027-40 from FY2026 value for each indicator (real GDP growth, current account, and fiscal balance).

[7] See Samoa Article IV consultation, 2017. The methodology used is consistent with the 2016 IMF Board Paper on "Small States' Resilience to Natural Disasters and Climate Change", which suggests that the long-term baseline macroeconomic projections for the countries highly vulnerable to natural disasters should not just be based on the outlook in non-disaster years, but should also factor in an average adverse impact from probable disasters to ensure more realistic medium-term and long-term plans and policies including focus on building resilience.

[8] "Samoa 2040: Transforming Samoa to a Higher Growth Path" is Samoa's national development agenda looking at key economic opportunities with the highest potential to achieve sustainable development by the year 2040. The document is based on seven thematic areas involving enhanced digital development, lifting tourism numbers and spending, greater labor mobility, boosting agriculture, effective investment in human capital, resilient development, and a safe and secure Samoa.

inflation to pick up but remain below 3 percent (meeting Central Bank of Samoa's indicative target rate),[9] with commodity prices remaining soft.

- **The current account balance** recorded a surplus of 1.2 percent of GDP in FY2020 driven by import compression (from collapse of the tourism sector, under-execution of CAPEX, and weak domestic demand), resilient remittance inflows, and a large increase in foreign grants in response to the COVID-19 pandemic. Moving forward, the pandemic and its aftermath are expected to continue weighing on tourism receipts and exports, and put pressure on remittance inflows. While additional official grant receipts will help soften both external and fiscal pressures, they will likely add to imports. The current account deficit is projected to widen to 6.5 percent and 10.9 percent of GDP in FY2021 and FY2022, respectively. Nonetheless, the deficit is expected to improve gradually, reaching 2.0 percent of GDP over the medium term, supported by tourism-related recovery. The current account deficit is higher in the long-term (FY2032-2041), assuming faster import growth driven by the ramping up of public investment. Nonetheless, it is expected that exports would also pick up over the long-term horizon, helping contain the current account deficit. To account for the average impact of natural disasters, the deficit is increased by 3.5 pp of GDP on average during FY2027-41.

- **The overall fiscal balance recorded a surplus of 6.2 percent of GDP in FY2020** but is projected to turn into a sizable deficit in FY2021-22, driven by the impact of COVID-19 on the tourism sector, considerable losses in associated government revenue, and lower grants receipts. Staff projects a deficit of 3.4 and 6.7 percent of GDP in FY2021 and FY2022, respectively. The widening fiscal deficit in FY2022 reflects lower grants (FY2020-21 are one-offs due to measles and COVID-19 shock) and a recovery in CAPEX implementation (FY2020 was very low at 3.5 percent of GDP). However, it is expected to improve to 3.0 percent of GDP on average over the medium-term (FY2023-26). The fiscal deficit in the last decade of the long-term projection horizon (FY2032-41) is assumed to be slightly larger than before, reflecting the government's plan to increase public investment projects, unlocking private investment. To account for the average annual impact of natural disasters, the deficit is widened by 3.5 pp of GDP on average during FY2027-41.

Text Table 7. Samoa: Baseline Macroeconomic Assumptions

(In percent of GDP, unless otherwise noted)

	Current DSA (2021 Article IV)				Previous DSA (RCF SR)			
	FY2021-26	FY2027-31	FY2032-41	FY2027-41	FY2020-2025	FY2027-2031	FY2032-41	FY2027-41
Real GDP growth (in percent change)	0.8	0.7	1.2	1.0	2.7	0.9	0.9	0.9
Inflation (in percent change)	1.7	2.6	2.8	2.7	2.6	2.8	2.8	2.8
Current account deficit	5.0	5.5	6.2	6.0	3.8	5.0	5.0	5.0
Overal fiscal deficit	3.7	4.8	6.3	5.8	6.0	4.5	5.7	5.2

Sources: Country authorities; and staff estimates and projections.

[9] The Central Bank of Samoa (CBS) relates Samoa's inflation rate to its major trading partners' annual average inflation rate—this determines the target that the CBS seeks to maintain each year—to pursue price stability. The indicative target rate has been 3 percent in recent years.

- **Continued eligibility for concessional borrowing from MDBs is assumed for the forecast period to finance the fiscal deficit, which is driven by the response to COVID-19 and increased infrastructure spending.** The planned new financing needs identified in the DSF are consistent with public gross financing needs in the baseline macro-economic framework. The baseline also incorporates cashflows (from delayed debt service) of around 2.3 percent of GDP[10] in FY2021 obtained in the context of the Debt Service Suspension Initiative (DSSI). All external financing is assumed to be concessional. In addition to IDA financing, 20 percent of the remaining financing need is assumed to be sourced from the AsDB while the remaining 80 percent is from other multilateral partners. The grant element of all new loans is 48 percent on average. It is assumed that borrowing from the IDA, the AsDB, and other Multilaterals is on full credit terms.[11]

9. The realism tools suggest that the macroeconomic and fiscal assumptions are reasonable.[12] Although the external and public PPG debt trajectory in the current DSA differs from the April 2020 DSA (Figure 3), this is explained by the lower debt outturn in FY2020 compared to a higher projection in the April 2020 DSA (46.8 percent of GDP compared to the projection of 55.6 percent of GDP). Most indicators underlying external debt dynamics remain broadly in line with their historical contributions except for the current account, which is highly volatile and vulnerable to external conditions. The underlying indicators for public PPG debt remain reasonable. The primary balance is projected to deteriorate between FY2021 and FY2022. The assumed revenue path discounts the collection of tax arrears in FY2020 and incorporates expected revenue losses from the impact of COVID-19 on the tourism sector amongst other things, and lower grants receipts. On the other hand, primary expenditure is projected to increase, driven mainly by capital projects including disaster-resilient infrastructure spending that supports growth. Staff real GDP growth projection during FY2021-22 is lower than the growth path projected by the realism tool, which is based on the fiscal adjustment[13] and a typical fiscal multiplier in a LIC (about 0.4).[14] Staff baseline projection is based on a prolonged impact of the COVID-19 pandemic (continues in FY2021 with a gradual recovery commencing FY2022). CAPEX implementation reflects border closures, local weather conditions, and a limited implementation capacity, thereby having contributed to under-execution of public investment projects (lower than budgeted) for the second consecutive year in FY2020, but it is expected to pick up gradually over the coming years (Figure 4).

[10] This includes savings from DSSI extension. The savings without the extension was around half of the total savings.

[11] With respect to projected new borrowing from the IDA and AsDB, DSAs always assume terms that would prevail if the country was at low risk of debt distress, independent of current actual terms (which can change on a year to year basis). This is done to avoid a circular situation where the assumption that future commitments will be on grant terms would yield actual commitments on credit terms.

[12] The negative residuals in FY2021 are associated with the liquidation of assets to cover financing needs. The authorities FY2021 Budget and staff projections assume no borrowing in FY2021 and the entire financing requirement is met by a drawdown of cash deposits (accumulated from the surpluses in FY2019 and FY2020).

[13] Fiscal adjustment of -10.1 percent (-3.9 in FY2021 minus 6.2 in FY2020) in FY2021 and -1.4 percent (-5.4 in FY2022 minus -3.9 in FY2021) in FY2022.

[14] Guidance Note on the Bank-Fund Debt Sustainability Framework for Low Income Countries (February 2018).

B. Standard Stress Test

10. There are six standardized stress tests in the DSF, each applied to both the external and the public DSA. These standard stress tests capture shocks to real GDP growth, the primary fiscal balance, exports, other flows (includes official and private transfers and FDI), exchange rate depreciation, and a combination of these shocks. In standardized stress tests, a variable is subject to a shock, and the post-shock values of the stressed variables are set to the baseline projection minus one standard deviation. However, if the historical average is lower than the baseline projection, then the post-shock values are instead set to the historical average minus one standard deviation.[15]

C. Tailored Stress Tests

Natural Disaster Stress Test (a medium-term scenario)

11. Given the severity and frequency of natural disasters in Samoa, a tailored stress test for a natural disaster shock in the medium term was also conducted to assess debt sustainability. Samoa is subject to a standard natural disaster shock which implies a one-off shock to public debt. This shock was proxied by the median size of damages caused by natural disasters in the past (equal to 21 percent of GDP) and adversely impacts GDP and exports growth in the second year of the projection period (FY2022 for this DSA). Based on staff research,[16] the default setting (10 percent of GDP increase in debt stock, 1.5 percent decline in real GDP growth, and 3 percent decline in exports growth) was modified so that the stress test captures Samoa-specific factors. The tailored stress test increases the public debt-to-GDP ratio by 21 percent in FY2022 – it assumes a natural disaster occurs in FY2022 and that new debt is contracted to finance damages, which are assumed to be in line with the median impact[17] of natural disaster in Samoa's history (the highest in the sample of 12 Pacific island countries using data from 1980-2016). Real GDP and exports growth decline by 3 percent and 6 percent in FY2022, respectively. Staff research also highlights that although the probability that at least one disaster occurs in a given year is relatively low compared to the sample, the probability per year for Samoa being exposed to a severe natural disaster is second highest in the sample.

Contingent Liability Stress Test

12. The stress test, capturing combined contingent liabilities, accounts for implicit liabilities and a potential financial market shock (Text Table 8). A contingent liability stress test reflects the likely consequences of a contingent liability shock on the debt path. The test incorporates contingent liabilities

[15] See 2018 Guidance Note on the Bank-Fund Debt Sustainability Framework for Low Income Countries.

[16] Detail in the IMF Working Paper 18/108, "The Economic Impact of Natural Disasters in Pacific Island Countries" (https://www.imf.org/en/Publications/WP/Issues/2018/05/10/The-Economic-Impact-of-Natural-Disasters-in-Pacific-Island-Countries-Adaptation-and-45826).

[17] The maximum damage Samoa has suffered from natural disasters is 161.8 percent of GDP while average damage amounted to 47.7 percent of GDP (Lee et al, 2018).

amounting to 7 percent of GDP, which comprises 2 percent of GDP of non-guaranteed SOE debt and 5 percent of GDP of financial market shock.

Text Table 8. Samoa: Combined Contingent Liability Shock

1	The country's coverage of public debt	The central government plus social security, government-guaranteed debt	
		Default	Used for the analysis
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2.0
4	PPP	35 percent of PPP stock	0.0
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0
	Total (2+3+4+5) (in percent of GDP)		7.0

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

DEBT SUSTAINABILITY

13. Samoa's mechanical risk rating is moderate for both external and overall debt. Judgement was applied to arrive at a high risk of debt distress by extending the projection horizon beyond ten years given there is a high probability of a large and protracted breach of the threshold under the baseline scenario over the long-run due to Samoa's exposure to frequent natural disasters and the effects of climate change.[18]

External Debt Sustainability Analysis

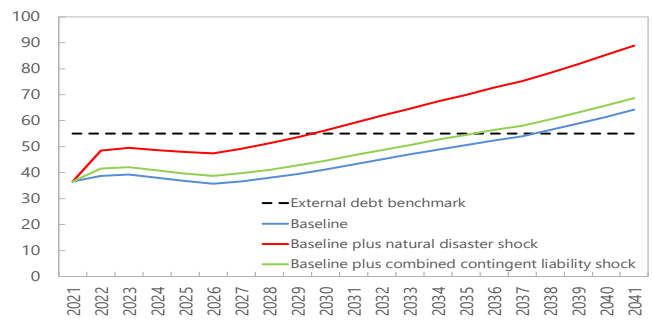
14. Under the baseline scenario, Samoa's external debt path is projected to breach the indicative threshold in the late 2030s (Figure 1).[19] The PV of the external debt-to-GDP ratio is expected to increase gradually from 37 percent of GDP in 2021 to 41 percent in 2030, and to reach 64 percent in 2041, on account of increased external borrowing to meet the fiscal financing need and finance new planned infrastructure projects as well as reconstruction projects resulting from future natural disasters. The PV of the external debt-to-GDP ratio breaches the indicative threshold of 55 percent of GDP in 2038 and remains above the threshold for the rest of the projection period (through 2041). Given a large share of concessional loans in external debt, the debt service-to-revenue ratio does not breach the indicative threshold.

[18] Paragraph 87 of the LIC DSF Staff Guidance Note (2018).

[19] The rating and application of judgement is in line with paragraph 87 of the LIC DSF Guidance Note (2018), which states that in exceptional circumstances, threshold breaches in years 11-20 may provide a rationale to change the risk rating. It is possible to consider a change in rating when (i) breaches are expected to be large, persistent, and thus resulting in significant differences relative to historical averages; and (ii) occur with a high probability despite occurring in the distant future. Such a situation could arise from trends that are not easily influenced by policy interventions, such as climate change (as in Samoa's case).

15. Stress tests confirm the vulnerability of external debt dynamics to shocks particularly exports, natural disasters, and contingent liabilities (Text Figure 1). All standard stress tests breach the threshold for the PV of external debt-GDP-ratio, with the exports shock causing the earliest breach (commencing 2032). This is a result of the lower base for exports (goods and services) in FY2021, reflecting the impact of COVID-19 on tourism earnings. For the tailored shocks, a natural disaster has the largest negative impact on the debt trajectory, causing a protracted breach to the PV of external debt-to-GDP ratio commencing in FY2030 (Figure 1). A tailored contingent liability shock breaches the PV of external debt-to-GDP ratio threshold from 2036 (Table 3).

Text Figure 1. Samoa: PV of External Debt-to-GDP Ratio
(Percent of GDP)

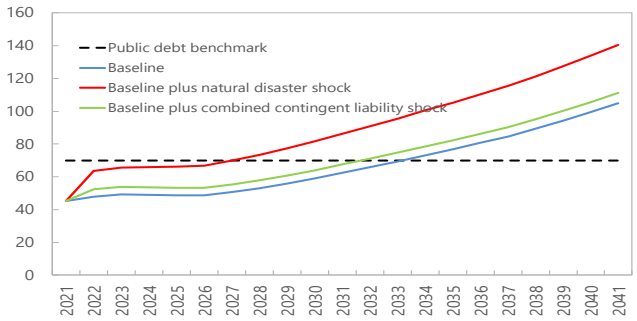


Source: IMF staff estimates

Public Sector Debt Sustainability Analysis

16. Total public sector debt closely follows external debt dynamics (Text Figure 2). Under the baseline scenario, the PV of public debt-to-GDP ratio breaches the benchmark in 2034 (Table 4). Most standard stress tests cause a breach to the benchmark level of the PV of public debt-to-GDP ratio, but the real GDP stress test causes an earlier breach in 2026. In addition, the natural disaster and contingent liability shocks result in threshold breaches in 2028 and 2032, respectively (Figure 2).

Text Figure 2. Samoa: PV of External Debt-to-GDP Ratio
(Percent of GDP)



Source: IMF staff estimates

RISK RATING AND VULNERABILITIES

17. Samoa's overall public debt situation remains fragile given its vulnerability to natural disasters and external shocks. PPG debt is expected to grow over the medium term. The debt-to-GDP ratio declined in FY2020, supported by a positive overall fiscal balance, which is considered temporary by staff given it is a byproduct of CAPEX under-execution and elevated grants from development partners. However, from FY2021, it is expected to increase, driven by a widening primary deficit, reflecting the impact of the COVID-19 pandemic. Samoa's overall public debt breaches the indicative threshold when the average annual impact of natural disasters is reflected in the long-term baseline

projections of GDP growth, the current account, and the fiscal balance. Also, a tailored natural disaster stress test results in a threshold breach earlier in the projection period.

18. Samoa remains at high-risk of debt distress. The PV of the external debt-to-GDP ratio and the PV of the public-debt-to-GDP ratio breach their indicative debt benchmark under the baseline scenario in 2038 and 2034, respectively. Another occurrence of a major natural disaster, external shock to exports, or a materialization of an implicit guarantees shock could substantially worsen the debt dynamics. In addition, applying a natural disaster stress test results in a protracted breach of the PV of public-debt-to-GDP ratio as early as 2028.

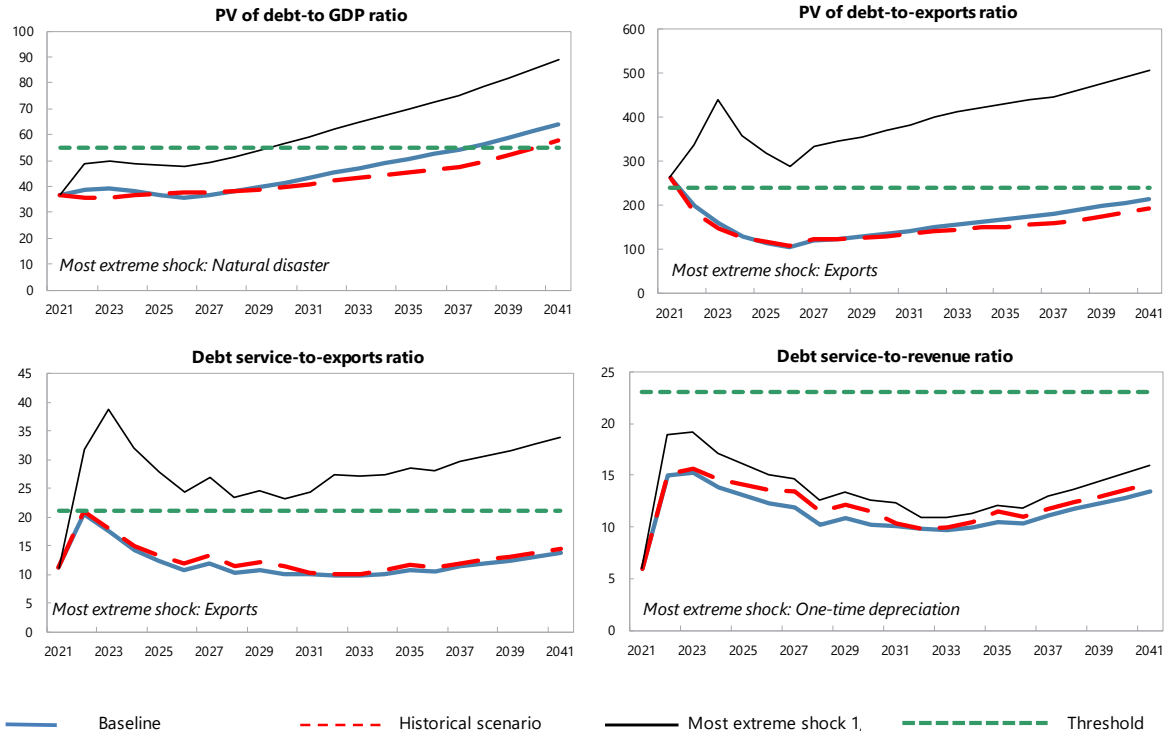
19. Samoa's debt sustainability hinges upon government efforts to build fiscal and external buffers and enhance resilience over the medium term. To meet the financing needs projected in staff baseline scenario, the authorities need to determine an appropriate financing mix, considering the importance of preserving their cash deposits for future "rainy days" and securing adequate financing on highly concessional terms. With the debt-to-GDP ratio breaching the authorities target of 50 percent of GDP by the end of the medium-term in the baseline scenario, additional budget support grants, revenue mobilization, spending prioritization, and PFM reforms are essential to ensure debt sustainability and pave the road for an inclusive economic recovery.

20. Well-timed, targeted, and temporary fiscal stimulus measures, if planned and executed properly, can help moderate the economic fallout of the COVID-19 pandemic. This would require running modest deficit levels with growth-promoting fiscal measures in the near term. Although growth promoting, inclusive policies (encompassing structural reforms) could raise public debt in the near term, they will help raise growth and fiscal revenues, building fiscal buffers and bringing the public debt trajectory downward over the medium term, improving long-term sustainability. Further support from the international community to help finance the deficit with the use of budget support grants and concessional loans will prove crucial in building adequate buffers and maintaining debt sustainability. It is critical for the authorities to make efforts to widen the coverage of the public-sector debt data by producing and disseminating aggregate data for extra-budgetary units, social security funds, and public corporations.

Authorities Views

21. The authorities agreed with staff's recommendation to remain sufficiently accommodative in the near term and showed strong commitment for medium-term consolidation to ensure debt sustainability. They plan to maintain an expansionary fiscal stance in the near term by extending temporary stimulus measures with better targeting and gradual withdrawal that will help stimulate the economy and ensure an inclusive, durable recovery. While an expansionary fiscal policy may breach the authorities' fiscal deficit target of 2 percent of GDP in the near term, they remain committed to fiscal consolidation over the medium term. The authorities agreed with staff's assessment that Samoa's debt is sustainable but remains at high risk of debt distress, given the country's high vulnerability to natural disasters.

Figure 1. Samoa: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2021-41 1/



Customization of Default Settings	Size	Interactions
Standardized Tests	Yes	
Tailored Stress		
Combined CL	No	
Natural disaster	Yes	Yes
Commodity price 2/	n.a.	n.a.
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	30	30
Avg. grace period	8	8

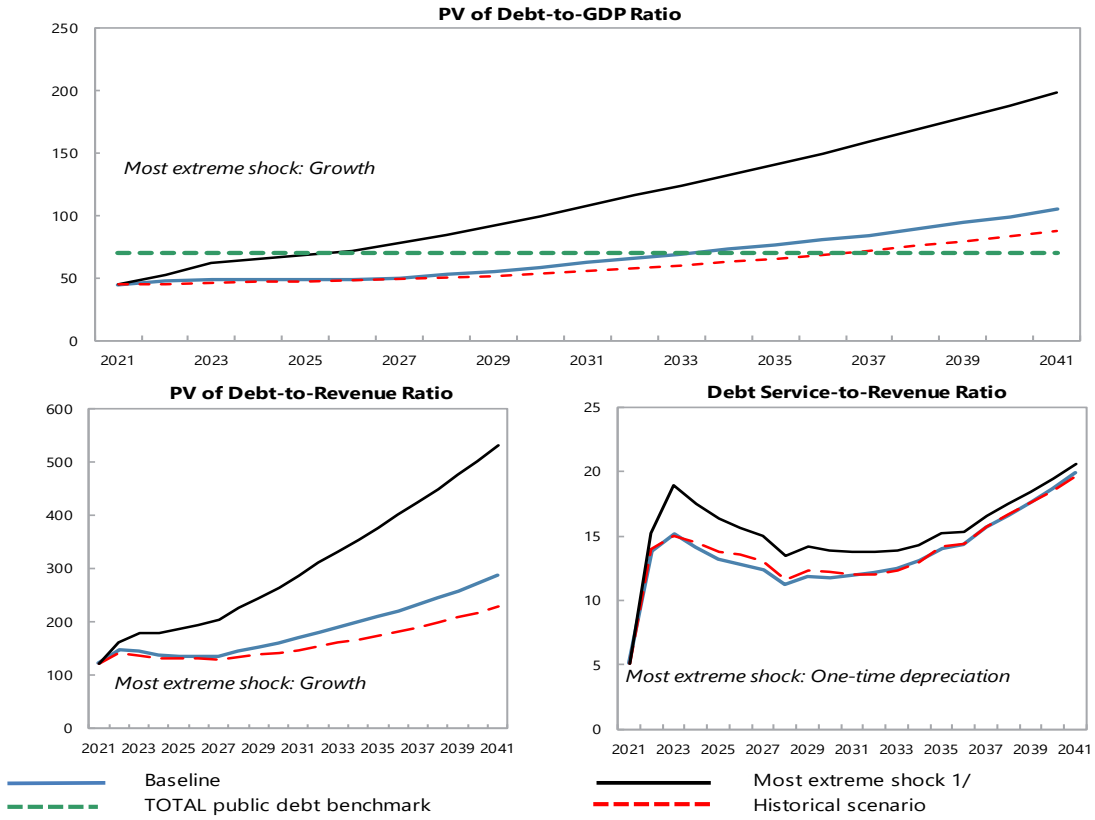
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Samoa: Indicators of Public Debt under Alternative Scenarios, 2021-41 1/



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	73%	73%
Domestic medium and long-term	25%	25%
Domestic short-term	1%	1%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
Avg. maturity (incl. grace period)	30	30
Avg. grace period	8	8
Domestic MLT debt		
Avg. real interest rate on new borrowing	8.6%	8.6%
Avg. maturity (incl. grace period)	40	40
Avg. grace period	30	30
Domestic short-term debt		
Avg. real interest rate	-2.1%	-2.1%

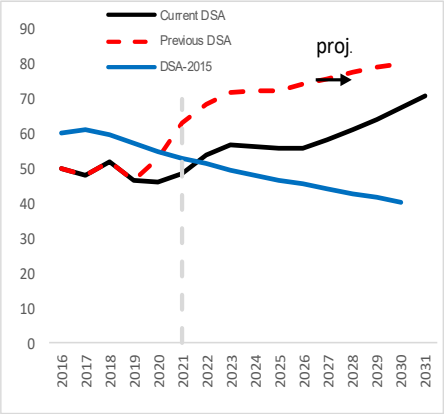
Sources: Country authorities; and staff estimates and projections.

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

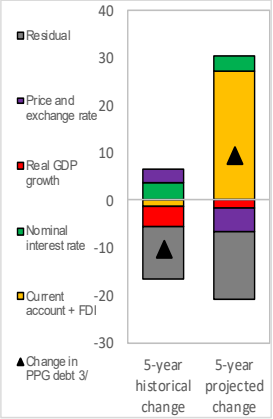
1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 3. Samoa: Drivers of Debt Dynamics – Baseline Scenario

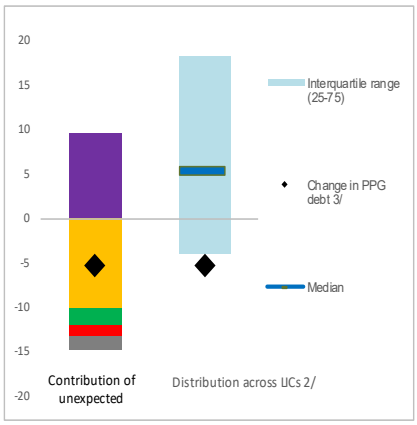
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

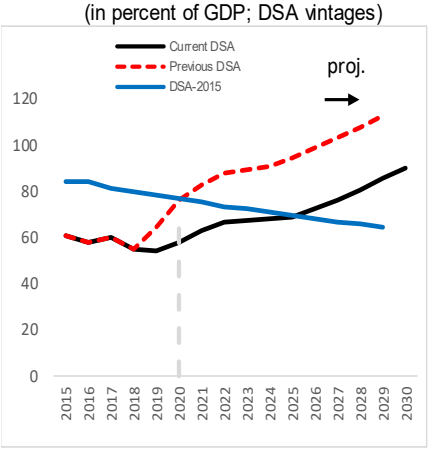


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

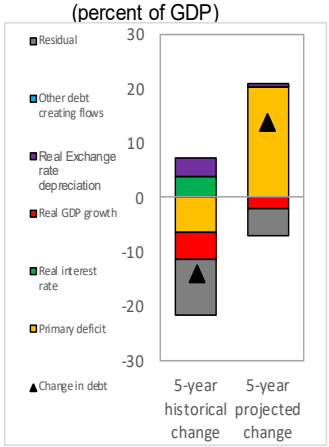


Public debt

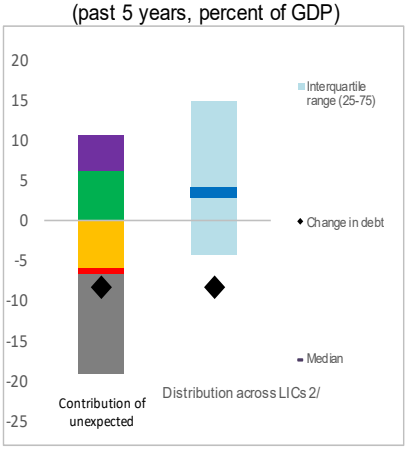
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



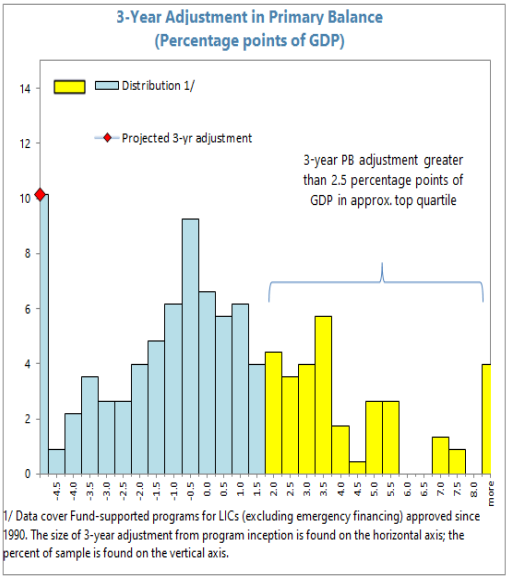
Sources:Country authorities; and staff estimates and projections

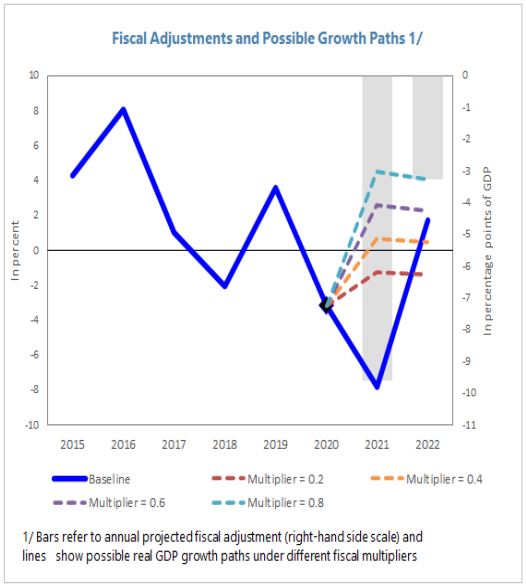
1/ Difference between anticipated and actual contributions on debt ratios.

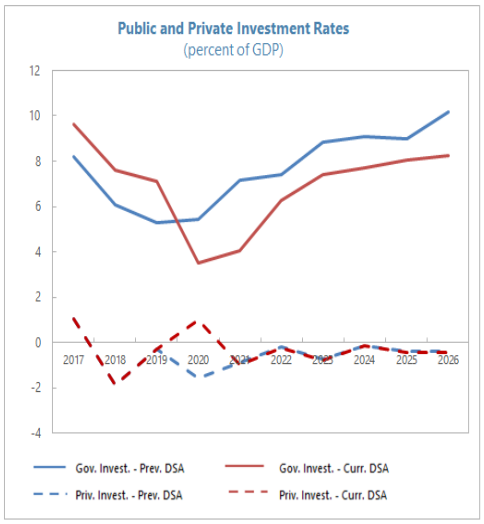
2/ Distribution across LICs for which LIC DSAs were produced.

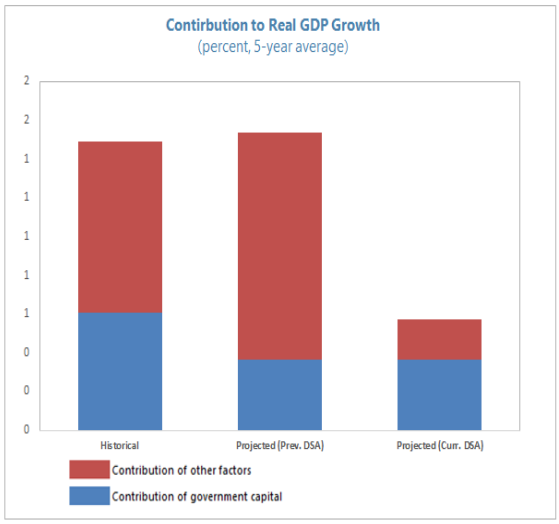
3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

Figure 4. Samoa: Realism Tools









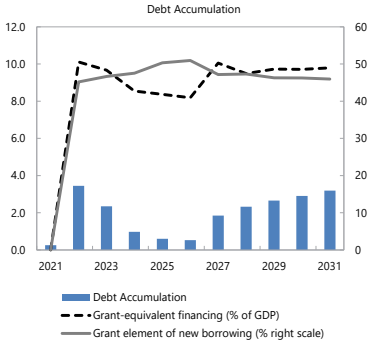
Sources: Country authorities; and staff estimates and projections.

Table 1. Samoa: External Debt Sustainability Framework, Baseline Scenario, 2018-41

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**51.9**	**46.8**	**46.3**	**48.7**	**53.9**	**56.6**	**56.3**	**55.9**	**55.6**	**70.9**	**102.7**	**50.0**	**59.0**
of which: public and publicly guaranteed (PPG)	**51.9**	**46.8**	**46.3**	**48.7**	**53.9**	**56.6**	**56.3**	**55.9**	**55.6**	**70.9**	**102.7**	**50.0**	**59.0**
Change in external debt	3.7	-5.1	-0.4	2.3	5.2	2.7	-0.3	-0.4	-0.3	3.5	3.9		
Identified net debt-creating flows	**0.8**	**-3.7**	**0.7**	**11.3**	**10.3**	**4.8**	**1.2**	**1.1**	**1.3**	**5.5**	**5.5**	**3.3**	**5.2**
Non-interest current account deficit	**-1.6**	**-3.7**	**-1.8**	**6.2**	**10.2**	**4.7**	**2.2**	**1.2**	**1.3**	**4.8**	**5.2**	**2.5**	**4.6**
Deficit in balance of goods and services	16.0	14.0	19.4	36.7	31.6	26.3	22.7	22.8	22.5	26.0	26.7	**20.8**	**26.6**
Exports	34.2	37.7	31.9	13.9	19.4	24.5	29.2	31.9	34.6	30.6	29.9		
Imports	50.2	51.7	51.3	50.6	51.0	50.8	51.9	54.8	57.0	56.5	56.5		
Net current transfers (negative = inflow)	-20.4	-21.3	-23.5	-32.5	-22.7	-22.8	-21.9	-23.0	-22.9	-22.9	-22.9	**-21.0**	**-23.7**
of which: official	-0.5	-0.7	-0.4	-6.6	-1.3	-1.3	-1.3	-1.2	-1.2	-1.2	-1.2		
Other current account flows (negative = net inflow)	2.8	3.5	2.3	2.0	1.3	1.3	1.3	1.5	1.8	1.8	1.4	**2.6**	**1.6**
Net FDI (negative = inflow)	**1.9**	**0.3**	**-1.0**	**1.0**	**0.2**	**0.8**	**0.1**	**0.4**	**0.4**	**0.4**	**0.4**	**1.0**	**0.5**
Endogenous debt dynamics 2/	**0.4**	**-0.3**	**3.5**	**4.2**	**-0.1**	**-0.8**	**-1.1**	**-0.6**	**-0.4**	**0.3**	**-0.1**		
Contribution from nominal interest rate	0.7	0.7	0.7	0.3	0.7	0.8	0.8	0.7	0.7	0.7	1.0		
Contribution from real GDP growth	1.0	-1.8	1.6	3.9	-0.8	-1.5	-1.9	-1.3	-1.1	-0.5	-1.1		
Contribution from price and exchange rate changes	-1.3	0.9	1.3	...	...	...	...	...	...	...	...		
Residual 3/	**2.9**	**-1.4**	**-1.1**	**-9.0**	**-5.1**	**-2.0**	**-1.5**	**-1.5**	**-1.7**	**-2.0**	**-1.6**	**-2.8**	**-3.0**
of which: exceptional financing	0.0	0.0	0.0	-2.3	0.0	0.0	0.1	0.1	0.1	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**34.0**	**36.7**	**38.8**	**39.2**	**38.0**	**36.8**	**35.8**	**43.2**	**64.2**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**106.8**	**263.6**	**199.4**	**160.1**	**130.2**	**115.3**	**103.6**	**141.5**	**214.9**		
PPG debt service-to-exports ratio	**8.7**	**7.9**	**10.9**	**11.2**	**20.6**	**17.6**	**14.1**	**12.3**	**10.7**	**10.0**	**13.7**		
PPG debt service-to-revenue ratio	**10.7**	**10.1**	**12.0**	**6.0**	**15.0**	**15.2**	**13.8**	**13.0**	**12.2**	**10.1**	**13.4**		
Gross external financing need (Million of U.S. dollars)	27.4	-3.4	5.3	65.4	112.5	80.1	55.4	50.7	51.5	89.7	148.6		
Key macroeconomic assumptions													
Real GDP growth (in percent)	-2.1	3.6	-3.2	-7.8	1.7	3.0	3.5	2.5	2.1	0.7	1.2	**1.1**	**0.8**
GDP deflator in US dollar terms (change in percent)	2.7	-1.7	-2.6	1.4	1.7	1.9	2.1	2.3	2.2	2.1	2.8	**0.9**	**2.0**
Effective interest rate (percent) 4/	1.5	1.3	1.3	0.7	1.4	1.5	1.4	1.4	1.3	1.1	1.1	**1.4**	**1.2**
Growth of exports of G&S (US dollar terms, in percent)	9.0	12.4	-20.3	-59.2	44.6	32.1	25.9	14.7	12.9	2.8	4.0	**3.6**	**6.6**
Growth of imports of G&S (US dollar terms, in percent)	7.1	4.9	-6.5	-7.8	4.4	4.4	8.1	10.6	8.6	2.8	4.0	**2.0**	**3.7**
Grant element of new public sector borrowing (in percent)	...	...	...	...	45.2	46.6	47.5	50.3	50.9	45.9	42.8	**...**	**47.4**
Government revenues (excluding grants, in percent of GDP)	27.7	29.7	29.1	25.8	26.8	28.3	29.8	30.2	30.3	30.3	30.3	**26.4**	**29.4**
Aid flows (in Million of US dollars) 5/	78.0	185.0	278.5	84.0	78.9	81.7	76.8	82.0	84.0	107.7	154.3		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	...	10.1	9.7	8.5	8.4	8.2	9.8	10.7	**...**	**9.4**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	...	64.5	67.3	72.8	75.7	76.9	70.2	63.4	**...**	**71.5**
Nominal GDP (Million of US dollars)	837	852	804	752	777	815	862	904	943	1,081	1,532		
Nominal dollar GDP growth	0.6	1.8	-5.7	-6.5	3.4	4.9	5.7	4.9	4.4	2.8	4.0	**2.0**	**2.8**
Memorandum items:													
PV of external debt 7/	...	...	34.0	36.7	38.8	39.2	38.0	36.8	35.8	43.2	64.2		
In percent of exports	...	...	106.8	263.6	199.4	160.1	130.2	115.3	103.6	141.5	214.9		
Total external debt service-to-exports ratio	8.7	7.9	10.9	11.2	20.6	17.6	14.1	12.3	10.7	10.0	13.7		
PV of PPG external debt (in Million of US dollars)			273.6	275.6	301.5	319.7	327.7	332.8	337.6	467.0	982.9		
(PVt-PVt-1)/GDPt-1 (in percent)				0.3	3.4	2.3	1.0	0.6	0.5	3.2	5.2		
Non-interest current account deficit that stabilizes debt ratio	-5.2	1.4	-1.4	3.9	5.0	2.0	2.5	1.6	1.7	1.3	1.3		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No



Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

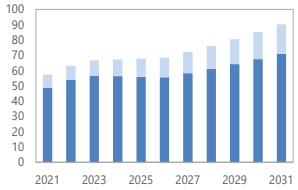
Table 2. Samoa: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018-41

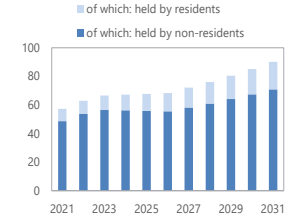
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**60.1**	**54.7**	**53.8**	**57.3**	**63.0**	**66.6**	**67.2**	**67.7**	**68.3**	**90.1**	**143.5**	**58.9**	**72.2**
of which: external debt	**51.9**	**46.8**	**46.3**	**48.7**	**53.9**	**56.6**	**56.3**	**55.9**	**55.6**	**70.9**	**102.7**	**50.0**	**59.0**
Change in public sector debt	2.5	-5.4	-1.0	3.5	5.7	3.7	0.6	0.5	0.6	5.0	6.7		
Identified debt-creating flows	**3.5**	**-3.7**	**-2.1**	**7.8**	**5.6**	**3.7**	**0.7**	**0.7**	**0.8**	**5.1**	**9.4**	**2.0**	**3.8**
Primary deficit	**-0.1**	**-2.7**	**-6.2**	**3.4**	**6.7**	**5.2**	**2.7**	**2.2**	**2.0**	**5.5**	**7.8**	**2.0**	**4.2**
Revenue and grants	32.1	35.7	38.5	37.0	32.3	33.9	35.5	35.9	36.0	36.6	36.4	**30.8**	**35.8**
of which: grants	4.4	6.0	9.4	11.2	5.5	5.6	5.6	5.6	5.6	6.3	6.1		
Primary (noninterest) expenditure	32.1	32.9	32.3	40.4	39.0	39.1	38.2	38.1	38.0	42.1	44.2	**32.8**	**40.0**
Automatic debt dynamics	**3.6**	**-1.0**	**4.1**	**4.4**	**-1.1**	**-1.5**	**-2.0**	**-1.6**	**-1.2**	**-0.4**	**1.6**		
Contribution from interest rate/growth differential	1.9	-1.7	2.3	4.4	-1.1	-1.5	-2.0	-1.6	-1.2	-0.4	1.6		
of which: contribution from average real interest rate	0.7	0.3	0.5	-0.2	-0.1	0.3	0.3	0.1	0.2	0.2	3.2		
of which: contribution from real GDP growth	1.2	-2.1	1.8	4.5	-1.0	-1.8	-2.2	-1.6	-1.4	-0.6	-1.6		
Contribution from real exchange rate depreciation	1.7	0.7	1.9	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-1.0**	**-1.7**	**1.1**	**-4.3**	**0.1**	**0.0**	**-0.1**	**-0.2**	**-0.1**	**-0.1**	**-2.7**	**-0.8**	**-0.4**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**41.2**	**45.3**	**47.9**	**49.2**	**49.0**	**48.6**	**48.6**	**62.4**	**105.0**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**107.1**	**122.4**	**148.2**	**145.2**	**137.9**	**135.6**	**135.1**	**170.6**	**288.5**		
Debt service-to-revenue and grants ratio 3/	**18.9**	**13.8**	**16.0**	**5.1**	**13.8**	**15.2**	**14.2**	**13.2**	**12.8**	**11.9**	**19.9**		
Gross financing need 4/	6.0	2.2	-0.1	5.3	11.1	10.3	7.7	7.0	6.6	9.8	15.0		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	-2.1	3.6	-3.2	-7.8	1.7	3.0	3.5	2.5	2.1	0.7	1.2	**1.1**	**0.8**
Average nominal interest rate on external debt (in percent)	1.5	1.3	1.3	0.7	1.4	1.5	1.4	1.4	1.3	1.1	1.1	**1.4**	**1.2**
Average real interest rate on domestic debt (in percent)	9.4	10.2	11.8	5.6	1.2	5.6	5.4	3.7	4.4	4.5	5.6	**4.9**	**4.3**
Real exchange rate depreciation (in percent, + indicates depreciation)	3.4	1.4	3.9	...	...	...	...	...	...	...	...	**0.7**	**...**
Inflation rate (GDP deflator, in percent)	2.0	2.2	0.3	-2.3	2.7	2.0	2.3	2.6	2.6	2.6	2.8	**1.5**	**2.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	-4.4	6.5	-5.1	15.4	-1.9	3.3	1.0	2.3	1.8	1.4	1.6	**2.3**	**3.3**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-2.6	2.7	-5.2	-0.1	1.0	1.5	2.1	1.7	1.4	0.5	1.1	**-1.7**	**0.9**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Samoa: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021-41

(In Percent)

	Projections 1/																				
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041
PV of debt-to GDP ratio																					
Baseline	37	39	39	38	37	36	37	38	40	41	43	45	47	49	51	52	54	**56**	**59**	**61**	**64**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2021-2031 2/	37	36	36	36	37	38	38	38	39	39	41	42	43	44	45	46	47	50	52	55	**58**
B. Bound Tests																					
B1. Real GDP growth	37	42	46	45	43	42	43	45	46	48	51	53	**55**	**57**	**59**	**62**	**63**	**66**	**69**	**72**	**75**
B2. Primary balance	37	40	43	41	40	39	40	42	43	45	47	49	51	53	55	**57**	**58**	**61**	**63**	**66**	**68**
B3. Exports	37	43	50	49	48	47	48	49	51	53	55	**56**	**58**	**59**	**60**	**61**	**63**	**64**	**66**	**68**	**71**
B4. Other flows 3/	37	42	46	45	43	42	43	45	46	48	50	52	53	55	**56**	**57**	**59**	**61**	**63**	**65**	**68**
B5. Depreciation	37	49	44	43	41	40	41	43	45	47	49	52	54	**57**	**59**	**62**	**64**	**68**	**71**	**74**	**78**
B6. Combination of B1-B5	37	45	49	48	47	46	47	48	50	52	54	**56**	**57**	**59**	**61**	**62**	**64**	**66**	**69**	**71**	**74**
C. Tailored Tests																					
C1. Combined contingent liabilities	37	42	42	41	40	39	40	41	43	45	47	49	51	53	54	**56**	**58**	**61**	**63**	**66**	**69**
C2. Natural disaster	37	48	50	49	48	48	49	51	54	**56**	**59**	**62**	**65**	**67**	**70**	**73**	**75**	**78**	**82**	**85**	**89**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio																					
Baseline	**264**	199	160	130	115	104	120	124	129	135	141	151	157	164	169	175	181	189	197	206	215
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2021-2031 2/	**264**	183	146	125	116	109	123	124	126	129	134	141	145	149	152	155	158	166	174	183	193
B. Bound Tests																					
B1. Real GDP growth	**264**	199	160	130	115	104	120	124	129	135	141	151	157	164	169	175	181	189	197	206	215
B2. Primary balance	**264**	208	174	142	126	114	132	137	142	148	155	165	171	178	183	190	195	203	211	220	229
B3. Exports	**264**	**336**	**440**	**360**	**320**	**289**	**335**	**345**	**357**	**370**	**383**	**402**	**412**	**422**	**430**	**439**	**448**	**461**	**475**	**490**	**506**
B4. Other flows 3/	**264**	217	187	153	136	122	142	146	152	158	164	173	178	183	188	193	197	204	211	219	227
B5. Depreciation	**264**	199	143	116	103	92	107	111	116	121	128	138	145	152	158	165	171	180	189	198	208
B6. Combination of B1-B5	**264**	**330**	183	**293**	**260**	235	**272**	**280**	**291**	**302**	**314**	**332**	**342**	**353**	**362**	**372**	**381**	**395**	**409**	**424**	**440**
C. Tailored Tests																					
C1. Combined contingent liabilities	**264**	214	172	140	124	112	130	135	140	146	153	163	170	177	182	189	195	203	212	221	230
C2. Natural disaster	**264**	**252**	205	169	152	139	163	170	178	186	196	210	219	229	237	**246**	**255**	**266**	**278**	**289**	**301**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio																					
Baseline	11	21	18	14	12	11	12	10	11	10	10	10	10	10	11	10	11	12	12	13	14
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2021-2031 2/	11	21	18	15	13	12	13	11	12	11	10	10	10	11	12	11	12	13	13	14	14
B. Bound Tests																					
B1. Real GDP growth	11	21	18	14	12	11	12	10	11	10	10	10	10	10	11	10	11	12	12	13	14
B2. Primary balance	11	21	18	14	13	11	12	10	11	10	11	11	11	11	12	11	12	13	13	14	15
B3. Exports	11	**32**	**39**	**32**	**28**	**24**	**27**	**23**	**24**	**23**	**24**	**27**	**27**	**27**	**29**	**28**	**30**	**31**	**32**	**33**	**34**
B4. Other flows 3/	11	21	18	15	13	11	12	11	11	11	11	12	12	12	12	12	13	13	14	14	15
B5. Depreciation	11	21	18	14	12	11	12	10	11	10	10	9	9	9	10	10	10	11	12	12	13
B6. Combination of B1-B5	11	**30**	**35**	**28**	**25**	**22**	**24**	21	**22**	20	**22**	**22**	**22**	**22**	**24**	**23**	**25**	**26**	**27**	**28**	**29**
C. Tailored Tests																					
C1. Combined contingent liabilities	11	21	18	14	12	11	12	10	11	10	10	10	10	10	11	11	12	12	13	13	14
C2. Natural disaster	11	**22**	19	15	13	12	13	11	12	11	11	11	11	11	12	12	13	13	14	15	15
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio																					
Baseline	6	15	15	14	13	12	12	10	11	10	10	10	10	10	11	10	11	12	12	13	13
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2021-2031 2/	6	15	16	15	14	14	13	12	12	11	10	10	10	10	11	11	12	12	13	14	14
B. Bound Tests																					
B1. Real GDP growth	6	16	18	16	15	14	14	12	13	12	12	11	11	12	12	12	13	14	14	15	16
B2. Primary balance	6	15	15	14	13	12	12	10	11	10	11	11	11	11	11	11	12	13	13	14	14
B3. Exports	6	15	16	15	14	13	13	11	12	11	12	13	13	13	13	13	14	14	15	15	16
B4. Other flows 3/	6	15	15	14	13	13	12	11	11	11	11	11	11	12	12	12	13	13	14	14	15
B5. Depreciation	6	19	19	17	16	15	15	13	13	13	12	11	11	11	12	12	13	14	14	15	16
B6. Combination of B1-B5	6	16	17	16	15	14	13	12	12	12	13	12	12	12	13	13	14	14	15	15	16
C. Tailored Tests																					
C1. Combined contingent liabilities	6	15	15	14	13	12	12	10	11	10	10	10	10	10	11	11	11	12	12	13	14
C2. Natural disaster	6	15	16	14	14	13	13	11	11	11	11	11	10	11	11	11	12	13	13	14	14
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.

2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

3/ Includes official and private transfers and FDI.

Table 4. Samoa: Sensitivity Analysis for Key Indicators of Public Debt, 2021-41

	Projections 1/																				
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041
PV of Debt-to-GDP Ratio																					
Baseline	45	48	49	49	49	49	51	53	56	59	62	66	69	**73**	**77**	**81**	**85**	**89**	**94**	**100**	**105**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2021-2031 2/	45	46	46	47	48	48	49	51	52	54	56	58	60	63	66	69	**72**	**76**	**79**	**83**	**88**
B. Bound Tests																					
B1. Real GDP growth	45	53	62	66	69	**72**	**78**	**85**	**92**	**100**	**108**	**116**	**125**	**133**	**141**	**150**	**159**	**168**	**178**	**188**	**199**
B2. Primary balance	45	51	55	54	54	54	56	59	62	65	69	**72**	**76**	**80**	**83**	**87**	**91**	**96**	**101**	**106**	**112**
B3. Exports	45	51	60	59	59	59	61	64	67	70	**73**	**76**	**79**	**82**	**86**	**89**	**92**	**96**	**101**	**106**	**111**
B4. Other flows 3/	45	51	56	56	55	55	57	60	63	66	69	**72**	**76**	**79**	**82**	**86**	**89**	**94**	**98**	**103**	**108**
B5. Depreciation	45	55	53	50	48	45	45	45	45	46	47	48	49	51	53	55	57	59	62	65	68
B6. Combination of B1-B5	45	51	54	54	54	55	58	61	64	68	**72**	**76**	**81**	**85**	**89**	**94**	**99**	**104**	**110**	**116**	**122**
C. Tailored Tests																					
C1. Combined contingent liabilities	45	52	54	54	53	53	55	58	61	64	68	**71**	**75**	**79**	**82**	**86**	**90**	**95**	**100**	**106**	**111**
C2. Natural disaster	45	64	66	66	66	67	70	**73**	**77**	**82**	**86**	**91**	**96**	**100**	**105**	**110**	**115**	**121**	**127**	**134**	**140**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio																					
Baseline	122	148	145	138	136	135	136	145	152	161	171	181	191	201	211	222	232	245	259	274	288
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2021-2031 2/	122	141	136	131	131	132	130	135	139	143	148	155	161	168	175	182	190	199	208	218	229
B. Bound Tests																					
B1. Real GDP growth	122	161	179	180	186	195	205	226	245	265	287	311	332	355	378	401	424	450	476	503	531
B2. Primary balance	122	157	161	153	151	151	152	161	169	178	188	198	208	219	229	240	251	264	278	292	307
B3. Exports	122	159	177	167	164	164	165	174	182	191	200	209	218	226	235	244	253	265	277	290	304
B4. Other flows 3/	122	159	165	157	154	153	154	163	171	180	189	199	207	217	226	236	245	258	270	284	298
B5. Depreciation	122	174	159	144	134	127	122	124	125	127	130	134	138	143	147	153	158	165	173	182	191
B6. Combination of B1-B5	122	157	158	151	150	151	153	164	173	184	195	207	219	231	243	255	268	283	299	315	332
C. Tailored Tests																					
C1. Combined contingent liabilities	122	162	159	151	148	148	149	158	166	175	185	196	206	216	226	237	248	262	275	290	305
C2. Natural disaster	122	196	192	185	183	185	188	200	210	222	235	248	261	275	288	302	316	332	349	366	384
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio																					
Baseline	5	14	15	14	13	13	12	11	12	12	12	12	12	13	14	14	16	17	18	19	20
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2021-2031 2/	5	14	15	14	14	14	13	12	12	12	12	12	12	13	14	14	16	17	18	19	20
B. Bound Tests																					
B1. Real GDP growth	5	15	18	17	16	16	16	16	17	17	18	19	20	22	24	25	28	30	32	34	36
B2. Primary balance	5	14	16	15	14	14	13	12	13	13	13	14	14	15	16	16	17	18	19	20	22
B3. Exports	5	14	15	15	14	13	13	12	12	12	13	14	15	15	16	16	18	19	19	20	22
B4. Other flows 3/	5	14	15	15	14	13	13	12	12	12	13	14	14	14	15	16	17	18	19	20	21
B5. Depreciation	5	15	19	18	16	16	15	13	14	14	14	14	14	14	15	15	17	18	18	19	21
B6. Combination of B1-B5	5	14	16	15	14	14	13	13	13	13	14	14	15	15	16	17	18	20	21	22	23
C. Tailored Tests																					
C1. Combined contingent liabilities	5	14	16	15	14	13	13	12	13	12	13	13	13	14	15	15	16	17	18	19	21
C2. Natural disaster	5	14	19	17	16	15	15	14	15	15	15	15	16	16	18	18	19	20	22	23	24
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.